FREEDOM INTERNATIONAL SCHOOL

WORKSHEET-MCQ

PHYSICS

CLASS XII

ELECTROMAGNETIC INDUCTION

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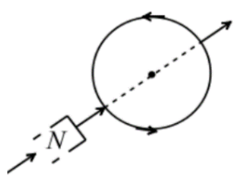
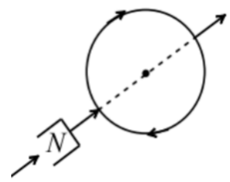
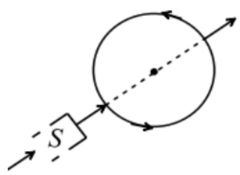
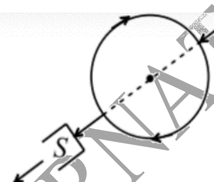
1. Which of the following figures correctly depicts the Lenz's law? The arrows show the movement of the labelled pole of a bar magnet into a closed circular loop and the arrows on the circle show the direction of the induced current

(a)


(b)


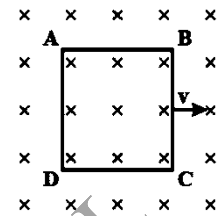
(c)


(d)


2. The magnetic flux linked with a coil at any instant t is given by $\Phi = 5\,t^3 - 100t + 300$ weber, the emf induced in the coil at t = 2 s is
   (a) -40 V (b) 40 V (c) 140 V (d) 300 V
3. If both the number of turns and core length of an inductor are doubled keeping other factors constant, self induction will be
   (a) unaffected (b) doubled (c) halved (d) quadrupled
4. In a coil when current changes from 10 A to 2 A in time 0.1 s, induced emf is 3.28 V. What is self-inductance of coil?
   (a) 4 H (b) 0.4 H (c) 0.04 H (d) 5 H
5. What is the coefficient of mutual induction, when the magnetic flux changes by 2×10^{-2} Wb and change in current is 0.01 A.
   (a) 2 H (b) 4 H (c) 3 H (d) 8 H
6. A wheel with 10 metallic spokes each 0.5 m long is rotated with a speed of 120 rev/min in a plane normal to the horizontal component of earth's magnetic field B_H at a place. If B_H= 0.4 G at the place, the magnitude of induced emf between the axle and the rim of the wheel is
   (a) 1.256×10^{-3} V (b) 6.28×10^{-4} V (c) 1.256×10^{-4} V (d) 6.28×10^{-5} V
7. A coil of cross- sectional area 400 cm^2 having 30 turns is making 1800 rev/ min in a magnetic field of 1 T. The peak value of the induced emf is

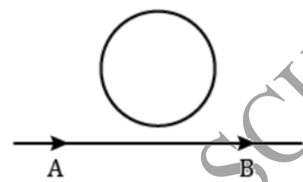
(a) 0.4 V (b) 0.6 V (c) 226 V (d) 2.26 V

8. A conducting square loop of side L and resistance R moves in its plane with a uniform velocity v perpendicular to one of its sides. A magnetic induction B constant in time and space, pointing perpendicular and into the plane of loop exists everywhere. The current induced in the loop is



(a) Blv/R clockwise (b) Blv/R anticlockwise
(c) 2 Blv/R anticlockwise (d) zero

9. In the arrangement shown in the given figure current from A to B is increasing in magnitude. Induced current in the loop will



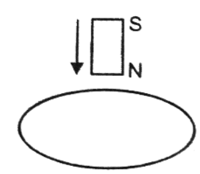
(a) have clockwise direction
(b) have anticlockwise direction
(c) be zero
(d) oscillate between clockwise and anticlockwise

10. The magnetic flux through a circuit of resistance R changes by an amount $\Delta\Phi$ in time Δt. Then the total charge q that passes during this time through any point of the circuit is given by
(a) $q= \Delta\Phi/ \Delta t$ (b) $q= \Delta\Phi\, R/ \Delta t$ (c) $q=-\Delta\Phi\, R/ \Delta t$ (d) $q= \Delta\Phi/ R$

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

11. **Assertion:** The bar magnet falling vertically along the axis of the horizontal coil will be having acceleration less than g.
Reason: Clockwise current is induced in the coil.



12. **Assertion:** Inductance coils are made of copper.
Reason: Induced current is more in wires having less resistance.
13. **Assertion:** Self-inductance is called the inertia of electricity.
Reason: Mutual induction is the phenomenon, according to which an opposing induced e.m.f. is produced in a coil as a result of change in current or magnetic flux linked in the coil.
14. **Assertion:** The induced emf in a conducting loop of wire will be non zero when it rotates in a uniform magnetic field.
Reason: The emf is induced due to change in magnetic flux.
15. **Assertion:** When two coils are wound on each other, the mutual induction between the coils in maximum.
Reason: Mutual induction does not depend on the orientation of the coils.